

MODEL QUESTION PAPER
PRECAST AND PRESTRESSED CONCRETE

PART A

I. Answer all questions in one word or sentence.

1. List any two structural precast element
2. List any two non structural precast element
3. High strength mix should have a water cement ratio between _____
4. High tensile bars threaded at ends are used in _____ system
5. Looping of high tensile tendons around the concrete is used in _____ system
6. Loss of prestress due to shrinkage in concrete depends upon _____
7. Loss of prestress due to creep of concrete is influenced by _____
8. Concentric tendons in a concrete induce _____ stresses
9. Eccentric tendons in a concrete induce _____ stresses

PART B

II. Answer any eight questions from the following, each question carries 3 marks.

1. List out the advantages and disadvantages of precast sections over cast in situ RCC sections
2. Explain about precast sheet piles
3. List out the Tests for Precast components as per BIS standards
4. List out any four precast elements used in building construction
5. Describe the main environmental measures of adopting prefabrication/modular construction
6. Distinguish between Pre tensioning and post tensioning
7. Explain Magnel-Blaton system of post tensioning
8. A pre stressed beam of size 300x450mm with span 4.5m is subjected to a udl of 15kN/m including the self weight of the beam. The prestressing tendons which are located along the longitudinal centroidal axis provide an effective prestressing force of 1000kN. Determine the extreme fibre stresses in concrete at the midspan section
9. Explain the various cable profiles in a rectangular prestressed beam with neat sketch
10. A simply supported prestressed concrete beam of rectangular cross section 300mm x 450mm is loaded with a total uniformly distributed load of 200kN over a span of 5m. It is subjected to a pre stressing force of 1500kN and tendons are in concentric. Sketch the distribution of stress at mid span and support.

PART C

Answer All questions. Each question carries 7 marks

III. Describe any three Structural Precast elements briefly

OR

IV. Describe Testing of Precast components as per BIS standards

V. Explain Modular co-ordination

OR

VI. Describe precast load bearing and non load bearing wall panels

VII. A pre tensioned concrete beam of rectangular section, 150mm breadth and 300mm deep is prestressed by eight high tensile wires of 7mm diameter located at 100mm from the soffit of beam. If the wires are tensioned to a stress of 1100N/mm^2 , calculate the percentage loss of stress due to elastic deformation assuming modulus of elasticity of concrete and steel as 31.5 kN/mm^2 and 210kN/mm^2

OR

VIII. Calculate the loss of stress in the tendons due to shrinkage of concrete in a post tensioned beam if the age of concrete at transfer is 10 days. Take $E_s = 2.10 \times 10^5$

IX. Calculate the loss of stress in the tendons due to shrinkage of concrete in a post tensioned beam if the age of concrete at transfer is 10 days. Take $E_s = 2.10 \times 10^5$

OR

X. Explain the steps involved in Design of simply supported rectangular beam section

XI. Explain how “green building” and precast concrete elements related

OR

XII. List out the limitations of prefabrications

XIII. Explain the need for the use of high strength concrete and tensile steel in Pre stressed concrete?

OR

XIV. A simply supported prestressed concrete beam of rectangular cross section 350mm x 550mm is loaded with a total uniformly distributed load of 250kN over a span of 6m. It is subjected to a pre stressing force of 2000kN and tendons are in concentric. Sketch the distribution of stress at mid span and support.